

Maintenance Documentation: Technical Overview

1. System Architecture

The application follows a **Client-Server architecture** using multi-threading to facilitate real-time communication.

- **Server Component:** The ChatServer acts as the central hub. It utilizes a ServerSocket to listen for incoming connections and spawns a new ClientHandler thread for every connected user. This ensures that the server can handle multiple clients without blocking.
- **Client Component:** The ChatUI maintains a persistent socket connection to the server. A background thread is continuously running to listen for incoming broadcast messages from the server.

2. Security Configuration

The application enforces security through two main mechanisms:

- **Authentication Handshake:**
 - **Mechanism:** A mandatory string match process.
 - **Location to Update:**
 - Server: ClientHandler.java (Line identifying the password comparison).
 - Client: ChatUI.java (The out.println line within connectToServer()).
 - **Instruction:** Both files must contain the exact same string to allow connection.
- **AES Encryption:**
 - **Mechanism:** Uses javax.crypto libraries to perform symmetric key encryption.
 - **Key Management:** The encryption key is hardcoded in EncryptionUtils.java as MySuperSecretKey.
 - **Updating the Key:** To increase security, replace the KEY variable in EncryptionUtils.java with a 16-character string. **Warning:** Every client and the server must use the identical key, or messages will fail to decrypt.

3. Communication Protocol

The data flow follows this sequence:

1. **Original Message** (Client Input)
2. **Encryption** (EncryptionUtils.encrypt())
3. **Network Transmission** (Socket out.println())
4. **Decryption & Re-encryption** (Server ClientHandler processes)
5. **Broadcasting** (Server sends to all clientWriters)
6. **Final Decryption** (ChatUI displays)

4. Troubleshooting & Maintenance

- **Network Port:** The application defaults to port 5000. If you encounter a `java.net.BindException`, the port is already in use. You can change this in both `ChatServer.java` and `ChatUI.java`.
- **Dependency Management:** This project relies on standard JDK libraries (`javax.swing`, `java.net`, `javax.crypto`). No external JARs are required.
- **Log Updates:** All connection attempts and errors are logged to the NetBeans **Output** window.